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Instruments and methods for tensioning a spinal tether

Abstract

Various methods and devices are provided for tensioning a tether. In one embodiment, a tether tensioning device is provided and includes an elongate shaft adapted to be positioned adjacent to a bone anchor implanted in bone, and a tensioning mechanism pivotally associated with the elongate shaft and adapted to couple to a tether seated across the bone anchor and to pivot relative to the elongate shaft to apply a tensioning force to the tether.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 17/076,380, filed on Oct. 21, 2020. U.S. application Ser. No. 17/076,380 is a continuation of U.S. application Ser. No. 16/229,094, filed on Dec. 21, 2018 (now U.S. Pat. No. 10,842,541). U.S. application Ser. No. 16/229,094 is a continuation of U.S. application Ser. No. 15/158,696, filed on May 19, 2016 (now U.S. Pat. No. 10,188,435). U.S. application Ser. No. 15/158,696 is a continuation of U.S. application Ser. No. 14/693,929, filed on Apr. 23, 2015 (now U.S. Pat. No. 9,370,390). U.S. application Ser. No. 14/693,929 is a continuation of U.S. application Ser. No. 11/842,376, filed on Aug. 21, 2007 (now U.S. Pat. No. 9,039,711). The entire contents of these applications are incorporated by reference herein.

FIELD

(1) The present application relates to devices and methods for tensioning a tether extending between bone anchors implanted in bone.

BACKGROUND

(2) Spinal deformities, which include rotation, angulation, and/or curvature of the spine, can result from various disorders, including, for example, scoliosis (abnormal curvature in the coronal plane of the spine), kyphosis (backward curvature of the spine), and spondylolisthesis (forward displacement of a lumbar vertebra). Early techniques for correcting such deformities utilized external devices that apply force to the spine in an attempt to reposition the vertebrae. These devices, however, resulted in severe restriction and in some cases immobility of the patient. Furthermore, current external braces have limited ability to correct the deformed spine and typically only prevent progression of the deformity. Thus, to avoid this need, several rod-based techniques were developed to span across multiple vertebrae and force the vertebrae into a desired orientation.

(3) In rod-based techniques, one or more rods are attached to the vertebrae at several fixation sites

to progressively correct the spinal deformity. The rods are typically pre-curved intraoperatively to a desired adjusted spinal curvature. Wires as well as bone screws can be used to pull individual vertebra toward the rod. Once the spine has been substantially corrected, the procedure typically requires fusion of the instrumented spinal segments.

(4) While several different rod-based systems have been developed, they tend to be cumbersome, requiring complicated surgical procedures with long operating times to achieve correction. Further, intraoperative adjustment of rod-based systems can be difficult and may result in loss of mechanical properties due to multiple bending operations. The rigidity and permanence of rigid rod-based systems can also hinder or prevent growth of the spine and they generally require fusion of many spine levels, drastically reducing the flexibility of the spine. To help remedy some of these issues, a tether and anchor system can be used to correct curvature of the spine using a number of anchors disposed within the spinal bones connected with tethers extending between them. The elasticity of the tethers prevents some of the problems with the rigidity and permanence of the rod-based systems, although the tethers must be tensioned after implantation to achieve the desired force between the anchor to correct the spinal deformities.

(5) Accordingly, there remains a need for improved methods and devices for correcting spinal deformities and, in particular, there remains a need for methods and devices for tensioning a tether extending between anchors implanted in bone.

SUMMARY

(6) Various methods and devices for tensioning a tether are provided. In one embodiment, a tether tensioning device is provided and includes an elongate shaft adapted to be positioned adjacent to a bone anchor implanted in bone. A tensioning mechanism can be pivotally associated with the elongate shaft and it can be adapted to couple to a tether seated across the bone anchor and to pivot past a longitudinal axis of the elongate shaft to apply a tensioning force to the tether. In particular, the tensioning mechanism can be adapted to move from a first position in which the tensioning mechanism is pivoted away from the elongate shaft, to a second position in which the tensioning mechanism is pivoted past the longitudinal axis of the elongate shaft to apply a tensioning force to a tether. In one exemplary embodiment, the elongate shaft can include an elongate slot formed therein. The elongate slot can be adapted to receive the tensioning mechanism therethrough to facilitate pivoting of the tensioning mechanism past the longitudinal axis of the elongate shaft. Various techniques can be used to mate a tether to the tensioning mechanism. For example, the tensioning mechanism can include one or more openings formed therethrough and adapted to receive a portion of a tether to allow the tether to removably couple to the tensioning mechanism. The device can also include a handle coupled to a proximal end of the elongate shaft.

(7) In one exemplary embodiment, the tensioning mechanism can also be adapted to move longitudinally along an axis of the elongate shaft between proximal and distal positions. The tether tensioning device can also include a biasing element coupled to the tensioning mechanism and adapted to bias the tensioning mechanism to the proximal position. For example, the biasing element can be a spring disposed within the elongate shaft and coupled to a distal portion of the tensioning mechanism to bias the tensioning mechanism proximally. The spring can be adapted to be compressed to allow the tensioning mechanism to move distally and to thereby control the amount of tensioning force applied to a tether as the tensioning mechanism is moved to the second position.

(8) In another embodiment, a tether tensioning device is provided and includes an elongate shaft having a tensioning mechanism pivotally associated with the elongate shaft and adapted to couple to a tether to apply a tensioning force thereto. The device can also include an overload mechanism coupled to the tensioning mechanism. The overload mechanism can be adapted to allow the tensioning mechanism to move distally along an axis of the elongate shaft when a threshold force is applied to the tether. In one embodiment, the tensioning mechanism can be pivotally coupled to a body that is slidably coupled to the elongate shaft, and the overload mechanism can be a biasing

element that biases the body proximally along the elongate shaft. In an exemplary embodiment, the overload mechanism can be in form of a spring disposed within the elongate shaft and coupled to a distal portion of the tensioning mechanism. The spring can be adapted to be compressed by the tensioning mechanism when a threshold force is applied to a tether coupled to the tensioning mechanism. In order to apply the tensioning force to the tether, the tensioning mechanism can be adapted to move from a first position in which the tensioning mechanism is pivoted away from the elongate shaft, to a second position in which the tensioning mechanism is pivoted past a longitudinal axis of the elongate shaft to apply a tensioning force to a tether. In one embodiment, the tensioning mechanism can include one or more openings formed therethrough for receiving a portion of the tether therethrough to allow the tether to removably couple to the tensioning mechanism.

(9) Methods for applying tension to a tether are also provided, and in one embodiment the method can include coupling a tether seated across a bone anchor implanted in bone to a tensioning mechanism pivotally associated with an elongate shaft, and pivoting the tensioning mechanism past a longitudinal axis of the elongate shaft to apply a tensioning force to the tether. Coupling the tether to the tensioning mechanism can include threading a portion of the tether through one or more openings formed in the tensioning mechanism. Pivoting the tensioning mechanism can include moving the tensioning mechanism from a first position in which the tensioning mechanism is pivoted away from the elongate shaft, to a second position in which the tensioning mechanism is pivoted past the longitudinal axis of the elongate shaft to apply the tensioning force to the tether. In one exemplary embodiment, the tensioning mechanism is pivoted past the longitudinal axis of the elongate shaft by pivoting into a slot formed in the elongate shaft. The tensioning mechanism can optionally be biased to a proximal position by a biasing element coupled to a distal end of the tensioning mechanism. When a threshold tensioning force is applied to the tether, the tensioning mechanism moves distally along a longitudinal axis of the elongate shaft. In one embodiment, a biasing element can bias the tensioning mechanism proximally, and the biasing element can be compressed as the tensioning mechanism moves distally.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various exemplary embodiments disclosed herein will be more fully understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view of one embodiment of a tether tensioning device having a tensioning mechanism shown in a first, untensioned position;
- (3) FIG. 2 is a perspective view of the tether tensioning device of FIG. 1 showing the tensioning mechanism in a second, tensioned position;
- (4) FIG. 3 is a front view of the tether tensioning device of FIG. 2;
- (5) FIG. 4 is a perspective view of the tether tensioning device of FIG. 2 showing a biasing element coupled to the tensioning mechanism;
- (6) FIG. 5 is a perspective view of the tensioning mechanism and biasing element of FIG. 4 showing a tether coupled to the tensioning mechanism that is in the first, untensioned position; and
- (7) FIG. 6 is a perspective view of the tensioning mechanism and biasing element of FIG. 4 showing a tether coupled to the tensioning mechanism that is in the second, tensioned position.

DETAILED DESCRIPTION

(8) Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein

and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present application.

(9) Various exemplary methods and devices are provided for tensioning a tether extending across one or more anchors implanted in bone. In general, a tensioning mechanism is provided that is adapted to couple to a tether seated across one or more bone anchors implanted in bone. The tensioning mechanism can be adapted to apply a tensioning force to the tether to allow the tether to be tensioned between the bone anchors.

(10) FIGS. 1-3 illustrate one exemplary embodiment of a tether tensioning device **10** that generally includes an elongate shaft **12** extending from a handle **14**. A tensioning mechanism **16** is pivotally associated with the elongate shaft **12** and it is adapted to mate to a tether **18** seated across one or more bone anchors implanted in bone. The tensioning mechanism **16** can be adapted to move from a first, untensioned position in which the tensioning mechanism **16** is pivoted away from the elongate shaft **12**, shown in FIG. 1, to a second, tensioned position in which the tensioning mechanism **16** is pivoted toward the elongate shaft **12** and past a longitudinal axis of the elongate shaft **12** to apply a tensioning force to the tether **18**. In one exemplary embodiment, the tensioning mechanism **16** can be pivoted into a slot **20** formed in the elongate shaft **12** to apply a tensioning force to the tether **18**, shown in FIG. 2. A person skilled in the art will appreciate that the tensioning mechanism **16** can be pivoted past the longitudinal axis of the elongate shaft **12** in a variety of ways, including but not limited to the tensioning mechanism **16** being moved around a side of the elongate shaft **12**, over an end of the elongate shaft **12**, or up into the elongate shaft **12**.

(11) The elongate shaft **12** can have a variety of configurations, and it can be flexible or rigid depending on the intended use. In an exemplary embodiment, the elongate shaft **12** has a length that allows the distal end **12d** of the elongate shaft **12** to be positioned adjacent to a tether **18** disposed along a spinal column, while the proximal end **12p** remains external to the body. This length allows the elongate shaft **12** to extend from the tissue surface to the treatment site, e.g., bone anchors implanted in vertebrae. A person skilled in the art will appreciate that the elongate shaft **12** can be made from a variety of biocompatible materials that have properties sufficient to enable the elongate shaft **12** to be inserted into the body. As further shown in FIGS. 1-3, the elongate shaft **12** can also include a slot **20** formed therein. The slot **20** can be configured to receive a portion of the tensioning mechanism **16** when the tensioning mechanism **16** is moved to the second position, as will be discussed in more detail below. Thus, the particular location and size of the slot **20** can vary depending on the location and size of the tensioning mechanism **16**. In the illustrated embodiment, the slot **20** is located in a proximal portion of the elongate shaft **12**, just proximal to the mating location between the tensioning mechanism **16** and the elongate shaft **12**, and it extends along about half or less than half of the length of the elongate shaft **12**. A person skilled in the art will appreciate that the slot **20** can have various configurations and it can be at a variety of locations on the elongate shaft **12**.

(12) The elongate shaft **12** can also include a handle **14** formed on or coupled to the proximal end **12p** of the elongate shaft **12**. The handle **14** can have any shape and size but it is preferably adapted to facilitate grasping and manipulation of the device **10**. As shown in FIG. 1, the handle **14** has a generally elongate cylindrical configuration and extends substantially perpendicular to the longitudinal axis of the elongate shaft **12**. A person skilled in the art will appreciate that any handle can be coupled to the elongate shaft, or that the device **10** can be used without a handle.

(13) As indicated above, the device **10** also includes a tensioning mechanism **16**, shown in more detail in FIGS. 4-6, that is associated with the elongate shaft **12** and that is adapted to apply a tensioning force to a tether **18**. The tensioning mechanism **16** can have a variety of shapes and sizes, but it is preferably adapted to grasp or mate to a tether **18** and to apply a tensioning force to the tether **18**. While the tensioning mechanism **16** can have a variety of configurations, in one

exemplary embodiment, as shown, the tensioning mechanism **16** is in the form of an arm that is pivotally coupled to the elongate shaft **12**. The tensioning mechanism **16** can be pivotally coupled to the elongate shaft in a variety of ways. For example, in the illustrated embodiment, a distal end **16d** of the tensioning mechanism **16** includes two legs that are pivotally coupled to first and second pivot bosses **24a**, **24b** disposed on the elongate shaft **12** at first and second pivot points **26a**, **26b**. The bosses **24a**, **24b** are located just distal to a distal end of the elongate slot **20**. A person skilled in the art will appreciate, however, that the tensioning mechanism **16** can be coupled to the elongate shaft **12** in any way that allows it to pivot with respect to the elongate shaft **12**. Moreover, a person skilled in the art will appreciate that the tensioning mechanism **16** can be removable from the elongate shaft **12**. The tensioning mechanism **16** can be pivotally attached to the elongate shaft **12** in a variety of ways, including nesting the tensioning mechanism **16** into the elongate shaft **12**, or coupling the tensioning mechanism **16** to the bosses **24a**, **24b** disposed on the elongate shaft **12**. In addition, the tensioning mechanism **16** can be positioned along the elongate shaft **12** at a variety of locations, but it is preferably positioned at a location that allows the tensioning mechanism **16** to be pivoted into the elongate slot **20**, and that allows the tensioning mechanism **16** to be manipulated by a user when the distal end of the elongate shaft **12** is positioned adjacent to a tether **18** implanted in the spine.

(14) The tensioning mechanism **16** can also include features to allow the tether **18** to be coupled thereto. In one embodiment, the tensioning mechanism **16** can include one or more openings **22** formed therethrough. The openings **22** can have any size and shape, but are preferably of a size and shape that facilitate threading the tether **18** therethrough. A person skilled in the art will appreciate that any features can be used to couple the tether **18** to the tensioning mechanism **16**, including any type of opening or grasping mechanism formed on the tensioning mechanism **16**. In one embodiment, where the tensioning mechanism **16** is removable from the elongate shaft **12**, the tether **18** can be pulled to the correct length by hand, and then coupled to the tensioning mechanism **16**, for example, through the openings **22**.

(15) In order to apply the tensioning force to the tether **18**, the tensioning mechanism **16** is adapted to pivot from a first, untensioned position in which the tensioning mechanism **16** is spaced apart from and extends transverse to the elongate shaft **12**, as shown in FIG. 1, to a second, tensioned position in which the tensioning mechanism is pivoted into the slot **20** to apply a tensioning force to a tether **18**. In particular, as the tensioning mechanism **16** is moved to the second position, the tether **18** is pulled proximally and then pulled through the slot **20**. In order to maintain the tensioning force applied to the tether **18** when the tensioning mechanism **16** is pivoted into the second, tensioned position, the tensioning mechanism **16** can be adapted to pivot in such a way as to allow at least a portion of the tensioning mechanism **16** to extend completely through the slot **20** formed in the elongate shaft **12**, as shown in FIG. 2. In other words, the tensioning mechanism **16** moves across the longitudinal axis and is positioned on an opposite side of the elongate shaft **12**. As a result, a portion of the tether **18** coupled to the tensioning mechanism **16** is positioned past the pivot points **26a**, **26b** and is being pulled with a downward force. This downward force maintains the tensioning mechanism **16** in the second, tensioned position, thus maintaining the tensioning force applied to the tether **18**. A person skilled in the art will appreciate that a variety of other techniques can be used to maintain the tensioning mechanism **16** in the second position, including various locking mechanisms.

(16) The tensioning mechanism **16** can also include features to control the amount of tensioning force that is applied to the tether **18**. In the illustrated embodiment, an overload mechanism is coupled to the tensioning mechanism **16** and it is adapted to allow the tensioning mechanism **16** to move distally along an axis of the elongate shaft **12** between a proximal position and distal position when a threshold force is applied to the tether **18**. This prevents a tensioning force greater than the threshold force from being applied to the tether **18**.

(17) The overload mechanism can have a variety of configurations, but in one embodiment it can

be in the form of a biasing element, such as a spring **28**, for biasing the tensioning mechanism **16** proximally. The spring **28** can be disposed in a channel **31** formed in the elongate shaft **12**, preferably at a position distal of the elongate slot **20**. A proximal end of the spring **28** can be coupled to a distal end of the tensioning mechanism **16**, and in particular the proximal end of the spring **28** can abut against a distal surface of a body **32** have the bosses **24a**, **24b** formed thereon. The body **32** can be movably disposed within the channel **31**. In order to maintain the body **32** and the spring **28** in the channel **31**, the channel **31** can include an elongate rod **34** disposed therein and extending through the spring **28** and through a bore **36** in the body **32**. This will allow the body **32**, and thus the tensioning mechanism **16**, to move proximally and distally within the channel **31**, thereby compressing and relieving the spring **28**. A distal end of the spring **28** can rest against a distal portion of the channel **31** formed in the elongate shaft **12**. In use, the biasing force of the spring **28** on the tensioning mechanism **16** can be overcome by pivoting the tensioning mechanism **16** until the tensioning force being applied to the tether **18** reaches the threshold force. This causes the tensioning mechanism **16** to move distally and thereby compress the spring **28**, thus preventing any additional tensioning force from being applied to the tether **18**. The tension of the spring **28** can be selected based on the desired threshold force. For example, a tightly-wound spring will yield a higher threshold force than a loosely-wound spring. A person skilled in the art will appreciate that a variety of other biasing elements can be used to control the amount of tensioning force being applied to the tether **18**.

(18) In use, the device **10** can be inserted through tissue, or through an access port disposed in tissue, so that the handle **14** and the tensioning mechanism **16** can extend from the tissue or port on the outside, and the distal end of the device **10** can be positioned adjacent to a bone anchor. The device **10** can be positioned adjacent to a tether **18** extending between bone anchors to be tensioning. For example, the device **10** can be removably mated to the bone anchor, or it can be positioned against a side of the bone anchor, or spaced apart from the bone anchor. A free end of the tether **18** can be grasped using a grasping device to allow the tether **18** to be removed from the body and coupled to the tensioning mechanism **18**, as shown in FIG. 5. Once the tether **18** is coupled to the tensioning mechanism **16**, the tensioning mechanism **16** can be pivoted towards the elongate shaft **12** and into and through the slot **20** to cause a tensioning force to be applied to the tether **18**, as shown in FIG. 6. If the tensioning force being applied to the tether **18** reaches a threshold force based on the tension of the spring **28** coupled to the tensioning mechanism **16**, the spring **28** will collapse, allowing the tensioning mechanism **16** to move distally along the elongate shaft **12**. This is effective to prevent the tensioning force being applied to the tether **18** from being greater than the threshold force.

(19) Once the proper tension is applied to the tether **18**, and the tether **18** has been locked to the bone anchor(s) to retain the tension thereon after the device **10** is removed, the device **10** can be removed. In one embodiment, the tether can be locked between adjacent bone anchors by applying a locking mechanism, such as a set screw, to each bone anchor. A person skilled in the art will appreciate that any mechanism can be used to lock the tensioned tether **18** between adjacent bone anchors.

(20) A person skilled in the art will appreciate that the various methods and devices disclosed herein can be formed from a variety of materials. Moreover, particular components can be implantable and in such embodiments the components can be formed from various biocompatible materials known in the art. Exemplary biocompatible materials include, by way of non-limiting example, composite plastic materials, biocompatible metals and alloys such as stainless steel, titanium, titanium alloys and cobalt-chromium alloys, and any other material that is biologically compatible and non-toxic to the human body.

(21) One skilled in the art will appreciate further features and advantages based on the above-described embodiments. Accordingly, the disclosure is not to be limited by what has been

particularly shown and described, except as indicated by the appended claims. All publications and references cited herein are expressly incorporated herein by reference in their entirety.

Claims

1. A tether tensioning device, comprising: an elongate shaft with a slot formed therein; and a tensioning mechanism comprising an arm pivotally associated with the elongate shaft, the arm adapted to couple to a tether seated across a bone anchor implanted in bone; wherein the arm is configured to pivot through the slot and past a central longitudinal axis of the elongate shaft to apply a tensioning force to the tether.
2. The device of claim 1, wherein the arm is adapted to move from a first position in which the arm is pivoted away from the elongate shaft, to a second position in which the arm is pivoted past the central longitudinal axis of the elongate shaft to apply a tensioning force to a tether.
3. The device of claim 1, wherein the arm is pivotally associated with the elongate shaft at a pivot point offset from the central longitudinal axis of the elongate shaft.
4. The device of claim 3, wherein the arm is configured to move a first portion of the tether from a first position in which the first portion of the tether is positioned at one side of the elongate shaft, to a second position in which the first portion of the tether is moved past the pivot point to an opposing side of the elongate shaft to apply a proximal force to a second portion of the tether.
5. The device of claim 4, wherein the arm is maintained in the second position by applying a distal force to the first portion of the tether.
6. The device of claim 1, wherein the arm includes one or more openings formed therethrough and adapted to receive a portion of a tether to allow the tether to removably couple to the arm.
7. The device of claim 1, wherein the tensioning mechanism is adapted to move longitudinally along an axis of the elongate shaft between proximal and distal positions.
8. The device of claim 7, further comprising a biasing element coupled to the tensioning mechanism and adapted to bias the tensioning mechanism to the proximal position.
9. The device of claim 8, wherein the biasing element comprises a spring disposed within the elongate shaft and coupled to the tensioning mechanism to allow the tensioning mechanism to move longitudinally along a portion of the elongate shaft.
10. The device of claim 9, wherein the spring is adapted to be compressed to control the amount of tensioning force applied to a tether as the tensioning mechanism is moved to the distal position.
11. A method of applying a tensioning force to a tether extending along a spinal column, comprising: mating a distal end of an elongate shaft to a bone anchor implanted in bone; coupling a tether seated across the bone anchor implanted in bone to a tensioning arm pivotally associated with the elongate shaft; and pivoting the tensioning arm through a slot formed in the elongate shaft and past a central longitudinal axis of the elongate shaft to apply a tensioning force to the tether.
12. The method of claim 11, wherein coupling the tether to the tensioning arm includes threading a portion of the tether through one or more openings formed in the tensioning arm.
13. The method of claim 11, wherein pivoting the tensioning arm includes moving the tensioning arm from a first position in which the tensioning arm is pivoted away from the elongate shaft, to a second position in which a first portion of the tether is moved past a pivot point disposed offset to the central longitudinal axis of the elongate body to apply a proximal force to a second portion of the tether.
14. The method of claim 13, wherein the tensioning arm is maintained in the second position by a distal force applied to the first portion of the tether.
15. The method of claim 11, wherein, when a threshold tensioning force is applied to the tether, the tensioning arm moves distally along a longitudinal axis of the elongate shaft.
16. The method of claim 15, wherein a biasing element biases the tensioning arm proximally, and the biasing element is compressed as the tensioning arm moves distally.

17. A method of applying a tensioning force to a tether extending along a spinal column, comprising: mating a distal end of an elongate shaft to a bone anchor implanted in bone; threading a tether seated across the bone anchor implanted in bone through one or more openings formed in a tensioning arm pivotally associated with the elongate shaft; and pivoting the tensioning arm past a central longitudinal axis of the elongate shaft to apply a tensioning force to the tether.
